

Modeling-Based Organizational Development: A Longitudinal Assessment*

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First-level supervisors have traditionally expressed frustration over their difficulty in effectively managing subordinates. This paper describes a social learning theory based on organizational development intervention focused on helping wood products plant supervisors develop better participative problem-solving skills. All the foremen in a plywood mill participated in workshops conducted over a three-month period. Measures of five key behaviors were taken immediately before the intervention began, shortly after it ended, and finally, six months later. Behavioral measures consisted of subordinates' perceptions of their supervisor. Results showed significant positive behavior change between before and after measures and the maintenance of this change for the six months thereafter. Significant improvement in productivity and labor relations indices also occurred.

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In the last two decades, Organization Development (OD) has emerged as one of the most widely used approaches for examining and implementing organizational change. Despite this popularity in both practice and discussion, relatively few planned attempts to change organizations have been rooted in a comprehensive theory of behavior. Perhaps as one consequence, OD has often had little impact on the long-term behavior of organizational members (Campbell, Dunnette, Lawler & Weick, 1970; Porras & Berg, 1978).

This article describes an organizational change project based upon social learning theory—a well-established and clearly articulated theory of human behavior (Bandura, 1977b; Mischel, 1973). Although social learning theory has proven an impressive guide for changing various behaviors in laboratory contexts, only recently have researchers employed this theory's approach to change organizational behavior. These applications mainly have utilized modeling based training programs to promote interpersonal skills in supervisory personnel (Goldstein & Sorcher, 1974; Kraut, 1976; Latham & Saari, 1979).

A full understanding of social learning theory's unique orientation toward the organizational change process requires a brief description of its major ideas. In this theory, behavior change is critically related to three classes of personal expectations. First, individuals are not likely to engage in a given course of behavior if they do not expect it to lead to desired outcomes (outcome expectations). Further, individuals will have differing values for any particular outcome (valence). A highly valued outcome (high positive valence) provides a

much stronger incentive to act than does one of low value.

As important as outcome expectations and valences are in determining action, they alone will not account for behavioral choices. Even though an individual may believe that a specific behavior will lead to a desired goal, he or she may judge her- or himself incapable of executing the necessary performance. The probability that an individual places on her or his ability to perform a particular behavior is termed efficacy expectation.

These three cognitive processes, in turn, are influenced by information gained through experience with the environment. Bandura (1977a) suggests that there are four types of experience responsible for changing expectations: performance accomplishments, vicarious experience, verbal persuasion, and physiological states. His research indicated that the first two—actually practicing a behavior or seeing another perform a behavior—are most likely to induce learning because they provide immediate and concrete guides about how to do something.

Vicarious experience alters behavior by changing expectations in two ways. First, it is influential because it assures the person that something can be done and provides a graphic picture of how to perform. Furthermore, if the individual perceives her- or himself similar to an observed model, expectations can be increased even more because he or she sees that the behavior is possible for someone like her- or himself (Meichenbaum, 1971). Secondly, by demonstrating what might happen as a result of behavior, potential outcomes for the observer's behaviors can be clarified (Bandura, Ross & Ross, 1966).

Personal accomplishment, of course, provides an even more powerful mechanism for change. Efficacy expectations are enhanced because the individual experiences her- or himself actually performing a behavior. Outcome expectations are clarified as an individual experiences, first hand, the consequences of action.

With these perspectives in mind, social learning theory's orientation toward planned organizational change efforts can be better understood by analyzing it in the context of OD's traditional theory, methods, and goals. For example, one major goal of most OD interventions is to enhance the quality of work life through attempts to improve interpersonal relationships (French & Bell, 1978; Huse, 1980). Current popular techniques for achieving this goal, such as team building, concentrate on increasing the expression of feelings and emotions among work team members in the context of task accomplishment. Frequently, this process requires the long-term presence of an outside consultant to guide members through "insight" experiences, with emphasis placed on changing the group or work team as a whole (French & Bell, 1978). In the above example, the social learning approach would suggest that insofar as team-building activities provide models for useful interpersonal skills and also allow for personal practice of these skills, then relationships among members of a work team may improve. Through influential modeling activities, efficacy and outcome expectations can be improved.

However, insofar as team-building or other OD interventions rely heavily on persuasion to convince individuals that group relations can be improved or merely attempt to change how in-

dividuals label their feelings about group interactions, then actual improvement in personal skills is likely to be less dramatic. To reiterate Bandura's (1977a) findings, these approaches are relatively less effective in altering the efficacy and outcome expectations that are crucial to behavior change.

A social-learning viewpoint also contributes to our perceptions of the consultant's proper role in OD. All change agents are concerned with producing permanent behavior change in the actual work setting. In typical OD practice, this goal has often been reached by relying on a continual presence of an outside consultant. Employing a social-learning approach makes this presence less necessary, as repeated rehearsal and a restructuring of work environment contingencies are used to strengthen personal skills enough so that an individual can be independent of the need for coaching from another.

OD practitioners following social learning theory concepts can also limit their presence in the change process in another way. By concentrating on altering specific behavioral skills rather than feelings about those skills, a program for concrete behavioral improvement can be developed instead of requiring that the individual gain insight about the reason for concern. If less effort is expended on helping the individual gain a complex understanding of her or his state, then efforts can be targeted toward creating very generalizable skills with a minimum of interpretation and direction on the part of consultants. Long-term change again is less dependent on the consultant's presence.

There is a final way in which social learning theory can contribute

to our understanding of OD intervention processes. Because social learning approaches emphasize change in concrete skills, practitioners should be more willing to describe, in concrete terms, the desired outcomes of a change program. Change agents must attempt to be clear about what specific capabilities will show that an intervention has achieved its goals. Too often, the goals of OD programs have been defined very vaguely in terms of enhancing broad abstractions such as self-actualization and emotional growth. Failure to describe such goals in terms of objective skills makes it difficult to determine whether an intervention has been successful. Such ambiguity also makes it more difficult for individuals to have confidence in their efficacy expectations. A social-learning approach emphasizes more readily observable guideposts of success.

There is an important distinction between social learning theory and the general expectancy theories that have achieved a significant status in the psychological literature (Heneman & Schwab, 1972; Irwin, 1971) and in the organizational literature (Porter & Lawler, 1968; Vroom, 1964). These theories and related empirical research focus primarily on action-outcome expectations and valences. Comparatively few efforts have been made to examine concepts related to efficacy expectations. In contrast, social learning theorists (Bandura, 1977a) have shown that efficacy expectations can be important predictors of behavior change.

The present research tested the utility of social-learning tenets as a guide for changing organizational behavior. As such, it built on the work of Latham and Saari (1979), who em-

ployed observational learning, or modeling, to create a successful program for teaching supervisory skills. In their study, 40 first-line supervisors were assigned to either a modeling-based program or a control condition. The modeling program was designed to improve supervisors' abilities to work with their employees in face-to-face interactions. Employing multiple internal and external criteria measured over time, results showed that the experimental group performed significantly better than did the control group on (a) behavioral role-play tests given three months later; (b) cognitive tests administered six months later; and (c) routine performance ratings administered one year later. In addition, supervisors who participated in the modeling program expressed highly favorable reactions to the program, both at the end of training and eight months later.

Latham and Saari (1979) focused their research on modeling as a method for developing competencies. Consequently, their measures concentrated on the extent to which supervisors acquired new skills and abilities. The research reported here extended this work by exploring the organizational impact of a modeling-based supervisory skill development program.

If, in fact, social learning theory provides an especially influential method for producing stable changes in supervisory behavior, then it is possible that real behavioral changes will also occur in their employees. In particular, employees may change their behavior because supervisors' actions provide a predictable and more rewarding set of outcome expectations for productivity. Therefore, it was predicted that providing supervisors with better interpersonal problem-solving

skills would ultimately result in changed employee attitudes, increased overall organizational performance demonstrated in greater worker productivity, and improvements in labor relations indices such as turnover, absenteeism, and grievance rates.

METHOD

This study was conducted in one experimental and two control plants of a multibillion-dollar, wood-products corporation. A thorough diagnosis of the plants indicated that major organizational problems existed between first-level supervisors and their subordinates. Efficient plant operation required close coordination of employee efforts to assure the smooth scheduling of product lines. The diagnosis indicated major breakdowns between employees and supervisors in communication and cooperation, resulting in lost productivity and high absenteeism.

Subjects

Seventeen first-line supervisors and approximately 700 of their union employees constituted the population of the experimental plant. Two additional plants from the same company, product line, and geographic region were used for comparison. One plant had 8 supervisors and approximately 350 union employees; the second had 5 supervisors and about 200 union employees.

Intervention process

Targeted supervisory skills were selected from a problem list created during the needs-assessment portion of the organizational diagnosis. These skills were grouped into 10 intervention

modules, each focusing on a specific problem activity. For example, one module helped supervisors deal with a person who was not performing up to standards; a second centered on absenteeism; a third was concerned with an employee who violated safety regulations; and a fourth dealt with ways to support valued employee behavior.

Although each of the 10 modules had a distinct focus, the intervention's overall goal was for supervisors to master a small set of interpersonal problem-solving skills and to effectively integrate their use. These skills and their underlying values were derived from principles of social learning theory (Bandura, 1977b; Mischel, 1973), communication theory (Rogers & Farson, 1955), and participative management theory (McGregor, 1960; Likert, 1967).

In the experimental condition, groups of 8 to 10 supervisors met in their plant for weekly half-day workshops over a period of 10 weeks. Typically, each session was led by a line manager, senior to the participants, who had undergone intensive skill development. The structure of each session was similar. First, a brief outline of the general principles underlying a module was presented, followed by a set of specific action points pertinent to the particular problem situation. A short videotape, modeling the desired behavior, was then shown. The videotapes assured identification by employing actual company personnel filmed in their own plants. Next, two supervisors rehearsed the behaviors modeled in the videotape. Supervisors were carefully guided by the workshop leader in their rehearsal to ensure that the requisite behaviors were practiced successfully.

Successful rehearsals were videotaped and observed by all participants. Discussion of positive aspects of performances allowed individuals to compare their behavior to that of the model and observe their own successful performance of the new behavior. This procedure was designed to enhance efficacy expectations.

To facilitate skill transfer, each participant then contracted to practice the skill on the job during the following week. The assignment was designed to ensure that supervisors practiced their skills and observed their effects on the job. In this manner, efficacy was further enhanced and outcomes clarified.

The following week's session began with each group member reporting the results of her or his contracts. A discussion of each person's experiences was used to reinforce supervisors for practicing target behaviors. Since the supervisors' superior was often the workshop leader, reinforcement during this phase was even more powerful.

Skill generalization was further emphasized in the final module of the program. Here, participants extended their newly acquired skills to new situations. They were required to use general principles learned from the other modules to create a new set of action steps aimed at dealing with a unique problem.

Intervention content

Component skills were embedded in a general problem-solving sequence: supervisors learned to describe a problem in a specific and nonjudgmental way (Behavior Description); explain the behavior's impact on organizational performance (Justification); elicit

and restate an employee's feeling about a problem (Active Listening); ask the employee for ideas regarding causes and solutions for a problem (Participative Problem Solving); and follow-up on agreed-upon solutions so that reinforcement could be given for improved performance (Follow-up and Reinforcement).

Measures

The intervention's impact on day-to-day supervisory behavior was of primary concern. Specifically, measures were employed to ascertain (a) whether changes in supervisory behavior were observed by employees, (b) whether these observations were accompanied by changes in employee attitudes and behaviors, and (c) whether these in turn were reflected in increased productivity.

To satisfy the first two goals, separate questionnaires, employing 7-point Likert scales, were developed for supervisors and employees as a means of evaluating use of the five basic skills. The employee questionnaire asked for ratings of specific behaviors that supervisors employed. In addition, employees were asked to relate feelings of their own status through questions designed to measure satisfaction, sense of involvement, and feelings of manipulation. Employees also responded to questions about the organization's general climate in terms of the flow of decision making, level of responsibility, and chances for advancement. Supervisors were asked to rate their own use of the five basic skills and to indicate their perceptions of the organization through similar questionnaires.¹ Finally, measures of productivity, turnover, absenteeism, and grievances were obtained from company records.

Questionnaires were administered to approximately one-third of the employees reporting to each supervisor (or a minimum of 10 employees). Employees were first stratified by supervisor and pay level (which reflected both seniority and job complexity) and then randomly selected to complete the questionnaire. Although plants were matched for comparability, assignment to the experimental condition was determined by the corporation based on its scheduling constraints.

Employee questionnaires were anonymous, except that a special code was used by each employee so that responses could be tracked over time. Both supervisors and employees completed their questionnaires during working hours. Supervisory responses, although identifiable by the researchers, were never seen by company officials.

Employing a nonequivalent control group design (Campbell & Stanley, 1963), questionnaires were administered one week before the 10-week intervention began (T1), one week after it ended (T2), and again six months after its conclusion (T3). Monthly performance and labor relations data were obtained from existing company records.

HYPOTHESES

Based on the conceptual scheme presented earlier, the following predictions were tested:

Hypothesis 1: *The increase in mean employee response (from T1 to T2 and T1 to T3) on questionnaire subscales measuring supervisors' use of (1) Behavior Description, (2) Justification, (3) Active Listening, (4)*

Participative Problem Solving, and (5) Reinforcement, would be significantly greater in experimental than in comparison plants.

Hypothesis 2: *The increase in mean employee response (from T1 to T2 and T1 to T3) on questionnaire subscales measuring their own (1) Satisfaction, (2) Sense of Involvement, and (3) Feelings of Manipulation, would be significantly greater in experimental than in comparison plants.*

Hypothesis 3: *The increase in mean employee response (from T1 to T2 and T1 to T3) on questionnaire subscales measuring the organization's overall climate would be significantly greater in experimental than in comparison plants.*

Hypothesis 4: *Labor relation decreases in (1) grievances, (2) absentee rate, and (3) turnover rate; and productivity increases in (1) production per labor man-hour, (2) recovery rate, and (3) average daily production (as obtained from company records) would be significantly greater in experimental than in comparison plants.*

RESULTS

Data were analyzed by calculating mean employee scores for each supervisor on each of the five behavioral skills. These mean employee ratings for each individual supervisor were then aggregated across supervisors, a process performed for both experimental and comparison conditions. Table 1 shows the mean employee ratings for experimental and comparison supervisors at each measure, for each of the five skills.²

Table 1. Supervisor Behavior Change
(A Comparison of Average Employee Ratings for Supervisors
in Experimental (E) & Comparison (C) Conditions)

Skill Variable	CON- DITION	T ₁	T ₂	T ₃	T ₃ -T ₁	tc,d	T ₃ -T ₁	t	T ₃ -T ₂	t
Behavior Description ^a	E	3.78	4.26	4.34	0.56	2.60***c	0.48	2.12***	0.08	0.66
	C	4.22	4.08	3.95	-0.27		-0.14	-0.14	-0.13	
Justification	E	3.39	4.05	4.04	0.65	2.09***	0.66	2.27***	-0.01	-0.55
	C	3.91	3.93	4.06	0.15		0.02	0.02	0.13	
Active Listening	E	4.14	4.65	4.73	0.59	1.91**	0.51	1.66**	0.08	0.46
	C	4.21	4.29	4.25	0.04		0.08	0.08	-0.04	
Participative Problem Solving	E	3.04	3.71	3.77	0.73	2.26**	0.67	1.29*	0.06	0.56
	C	3.14	3.49	3.40	0.26		0.35	0.35	-0.09	
Positive Reinforcement	E	3.31	3.95	4.03	0.72	1.56*	0.64	2.00**	0.08	-0.11
	C	3.42	3.57	3.69	0.28		0.15	0.15	0.12	

^aAll scales were designed *a priori* and found to be reliable. (Cronbach's Alpha .71 or greater)

^bN_{Experimental}=17
N_{Comparison}=13

cOne-Tail Test
ddf_E=16
df_C=12

*p≤0.09
**p≤0.05
***p≤0.01

Table 2. Organizational and Individual Attitude Change
(Experimental (E) vs. Comparison (C))

Variable	CON- DITION	T ₁	T ₂	T ₃	T ₃ -T ₁	t ^b	T ₂ -T ₁	t	T ₃ -T ₂	t
Satisfaction ^c	E ^a	4.41	4.36	4.48	0.07	0.08	-0.05	-0.65	0.12	0.78
	C	4.31	4.42	4.36	0.05		0.11		0.06	
Involvement	E	5.81	5.78	5.93	0.12	-0.02	-0.03	-1.14	0.15	1.16
	C	5.79	6.00	5.91	0.12		0.21		-0.09	
Manipulation	E	3.58	3.17	2.97	-0.61	-1.90**	-0.41	-1.49*	-0.20	-0.59
	C	3.43	3.37	3.27	-0.15		-0.05		-0.10	
Organizational Climate	E	3.43	3.73	4.02	0.59	2.32***	0.30	1.38*	0.29	1.36*
	C	3.76	3.84	3.87	0.11		0.08		0.02	

^aN_{Experimental} = 17
N_{Comparison} = 13

^bOne-Tail Test
df_E = 16
df_C = 12

^cAll scales were designed *a priori* and found to be reliable.
(Cronbach's Alpha .71 or greater)

* $p \leq 0.09$
** $p \leq 0.05$
*** $p \leq 0.01$

Hypothesis 1

Changes in supervisory behavior were assessed by making short-term (T1 to T2) and long-term (T1 to T3) comparisons of employee ratings. In addition, any maintenance or decay of target behaviors was determined by comparing employees' evaluations at T2 and T3. Experimental and comparison groups of supervisors were compared on these three change scores using t-tests.³

Experimental subjects showed significantly greater long-term change (T1 to T3) than did comparisons on each of the five variables. Most of the change occurred during the treatment phase (T1 to T2). During this time, the experimental group significantly improved on each of the five variables.

Increases in the experimental condition were sustained for each of the variables from T2 to T3, i.e., no significant decay in behavior change occurred during the six months after termination of all change activities.

Hypothesis 2

Analyses of change in employee attitudes were based on the same three comparisons reported above (see Table 2). Relative to the comparison group, employees in the experimental condition did not significantly increase their ratings of satisfaction or involvement over any of the time periods. Counter to the hypothesis, however, the experimental employees' perceptions of being manipulated by their supervisors actually *decreased* signifi-

Table 3. Regression Coefficients For Performance And Labor Relations Indices

Variable		B _E	B _C	t ^{a,b}
Labor Relations Indices	Grievances	-2.917 E-01	-5.83 E-01	1.04
	Absentee Rate	-17.920 E-01	-0.125 E-01	3.76****
	Turnover Rate	-4.917 E-01	37.50 E-01	14.02****
Performance Indices	Production/Direct Labor Man-Hour	7.330 E-03	-3.79 E-03	2.49**
	Recovery	1.158 E-01	-0.317 E-01	1.73**
	Average Daily Production	2.750 E+01	-0.596 E+01	1.47*

Note. Regression model is $Y = B_1 + B_2D + U$ where:

D=Dummy variable for the intervention

D=1 After Intervention

D=0 Before Intervention

u=error term

^aT-statistic calculated by: $t = (B_E - B_C) / \text{Standard Error of } B_E$

^bOne-tail Test

* $p \leq 0.05$

** $p \leq 0.025$

**** $p \leq 0.0025$

cantly from T1 to T3 relative to their comparison counterparts. Most of this change occurred during the treatment phase, but continued over the follow-up period.

Hypothesis 3

Employees in the experimental condition significantly increased their ratings of organizational climate relative to the comparison group over each of the time periods (see Table 2). Organizational climate was the only variable to improve significantly during the follow-up (T2 to T3).

Hypothesis 4

Performance and labor relations data are shown in Table 3. A regression analysis was performed using monthly data for each of the six months before the intervention for one part of the regression and the six months after the intervention for the second part. This method of analysis takes into account changes in trends of performance in the pre- and postintervention periods. The regression model accounted for these two periods through the use of a dummy variable. Separate regressions were computed for the experimental and comparison plants. Beta weights were compared to determine if the experimental group differed significantly from comparisons. Thus, this procedure tested for significant differences in the slope of regression lines that fit the productivity trends in the experimental and comparison plants.

The results indicate that, over the measurement periods, all three performance measures significantly improved in the experimental condition relative to the comparison. Absentee

rate and turnover rate showed similar significant improvements; grievance rate, however, did not.

DISCUSSION

The modeling intervention reported here was designed not only to improve supervisory skills, but to ensure that these skills were used on the shop floor where they could have a significant impact on employees' behavior and organizational performance. Employee perceptions indicated that supervisors did master the skills and use them in face-to-face interactions. Overall, changes were obtained not only in the target skills, but also in organizational climate, productivity, and indices of labor relations.

These results significantly extend the work of Latham and Saari (1979) who simply showed that a modeling-based supervisory program can change certain facets of supervisory behavior. The present research indicated that employees perceived changes in targeted supervisory skills and that these changes were influential enough to motivate greater productivity and improve feelings about the organization as a whole. Furthermore, these findings provide support for the idea that programs inspired by social learning theory can instill skills that are functional in a wide variety of settings and interactions.

Even though this social learning approach was particularly successful in developing a new overall supervisory orientation in participants, certain of the skills were observed to change more than others. For example, participative problem solving and positive reinforcement exhibited the greatest change. Whether the differen-

tial changes were due to differences in the supervisors' use of the skills or to differences in the employees' sensitivity to their use, remains to be examined. It may also be that these skills vary in their complexity and thus in their potential for development.

Anecdotal and interview data revealed further interesting consequences of the program. For instance, supervisory self-report data indicated satisfaction with the new skills. These data revealed that behavior change was a source of positive self-evaluation. In addition, supervisors in general were highly satisfied with the outcomes achieved by skill use.

On the other hand, employee feelings of satisfaction and involvement did not change. It may be that the instrument used to measure these factors was insufficiently sensitive. Satisfaction measures included global questions unrelated in a very direct way to supervisory change. Satisfaction was assessed in terms of peer relationship, pay and promotion, and other factors that extended beyond the immediate purpose of the intervention.

The measure of involvement used was perhaps similarly broad. It was assumed that the program would positively impact overall employee perceptions of the attractiveness of the working environment and thus their feelings of involvement with it. As it turned out, the program happened to be implemented at the same time as a major recession in the wood products industry. This event or other extraneous factors could have exerted an influence on global ratings of involvement (as well as the satisfaction measures described above).

Further research should include questions that ask about satisfaction and involvement with elements more specifically related to immediate supervisory performance.

It had been predicted that employees would feel more manipulated since supervisors learned a process that, if used inappropriately, could appear to be mechanical or artificial. Instead, employees' sense of being manipulated decreased. It appears, therefore, that even in the worst case of perceived mechanical use of the skills, the new behaviors were not judged as manipulative and in fact may well have seemed rewarding in comparison to supervisors' previous behaviors. Informal interviews supported this notion by revealing that the new skills were seen as a great improvement over old supervisory methods.

One of the most significant findings in this research was the supervisors' lack of regression to pre-intervention behavior levels six months after the end of the workshops. Few studies of planned change efforts have reported no behavioral loss after the end of the active intervention phase. This maintenance of skill levels is even more surprising in that no extensive intervention activity occurred following the workshop sessions.

It is also interesting to note that while skill levels did not fall after the formal development program, neither did they increase. This result is in spite of the fact that efforts were made to create generalizable skills that could be applied in an increasing variety of situations with a concomitant improvement in behavior. These findings contrast with the thrust of most OD interventions, i.e., the creation of a self-sustaining growth proc-

ess. It is possible that sufficiently strong efforts were not made to arrange positive contingencies in the actual work setting to support expanded skill performance. It is also possible that modeling-based technologies are most appropriate for "step-function" type change and may not produce a long-term growth process. This phenomenon certainly deserves further investigation.

In contrast to the behavior change reported above, attitudes about the organizational climate as a whole did improve after the formal intervention phase. The reason for these different development patterns for attitudes and behaviors is not totally clear. It may be that it takes some time for effects of supervisory behavior change to suffuse throughout the many different organizational groups, thereby enhancing the overall work climate. It makes sense that employees would have to be exposed to repeated examples of better supervision in multiple contexts before making the attribution of a better organization environment.

It may also be that climate change was induced by the method of program implementation. Requiring supervisors to practice skills within their natural work setting indicated management's commitment to real on-the-job changes. Having line managers, immediately superior to supervisors, serve as workshop leaders also indicated a strong concern for making the development program an everyday reality. Further, the very fact that the videotapes were filmed within the company and employed company personnel indicated still more company concern for creating the appropriate climate for permanent program acceptance. These are factors that any intervention should address to create

a long-term organizational impact, and it seems that social learning theory may guide practitioners to be more clearly attuned to the operation of these elements.

Finally, social learning theory need not be seen as a contrast to current OD theory. In fact, it could become a useful framework for the integration of many currently disparate perspectives and practices in OD. For example, it has the potential for providing a means to determine the utility and situational appropriateness of various intervention approaches. Specific OD techniques could be categorized according to how well they conform to Bandura's (1977b) hierarchy of influential experiences. Change orientations could be analyzed by determining what efficacy and outcome expectations they may alter. Practitioners would then be able to make more rational decisions about the most effective interventions for achieving their goals.

In summary, this research utilized a promising new organizational change strategy—modeling-based skill development drawn from the tenets of social learning theory. It produced significant improvement in supervisory behaviors, organizational climate, productivity, and indices of labor relations. As could be expected, however, many new questions were raised by the investigations. Future research needs to address such questions as (a) why some supervisors improved more than others; (b) the mechanism whereby supervisor behavior change affected organizational performance; (c) what role external reinforcers played in the acquisition and maintenance of new supervisory behaviors; (d) what impact the use of supervisors' superiors as workshop leaders had on behavior

change; (e) what kinds of skills are most useful and intrinsically rewarding to supervisors; and finally, (f) what changes will endure. As these questions are answered, perhaps a more unifying theory of planned organizational change will emerge.

NOTES

1. Scales used in this analysis were tested for reliability. Cronbach's ranged from .71 to .93.
2. For the t-tests of significance, the aggregation process at the supervisory level resulted in 16 degrees of freedom for the experimental condition and 12 degrees of freedom for the control condition.
3. Supervisor self-rating patterns of change were similar to those reported by employees in Table 1 except that, in all cases, supervisor self-ratings showed larger changes than employee ratings.

REFERENCES

- Bandura, A. Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 1977, 84, 197-215. (a)
- Bandura, A. *Social learning theory*. Englewood Cliffs, N.J.: Prentice-Hall, 1977. (b)
- Bandura, A., Ross, D., & Ross, S. A comparative test of the status envy, social power, and secondary reinforcement theories of identificatory learning. *Journal of Abnormal and Social Psychology*, 1966, 67, 527-534d.
- Campbell, J. P., Dunnette, M. D., Lawler, E. E., & Weick, K. E. *Managerial behavior, performance, and effectiveness*. New York: McGraw-Hill, 1970.
- Campbell, D. T., & Stanley, J. C. Experimental and quasi-experimental designs for research in teaching. In N. L. Gage (Ed.), *Handbook of research in teaching*. Chicago: Rand McNally, 1963. Pp. 171-246.
- French, W. L., & Bell, C. H. *Organization development* (2nd ed.). Englewood Cliffs, N.J.: Prentice-Hall, 1978.
- Goldstein, A., & Sorcher, M. *Changing supervisory behavior*. New York: Pergamon, 1974.
- Heneman, H. G., III, & Schwab, D. P. Evaluation of research on expectancy predictions of employee performance. *Psychological Bulletin*, 1972, 78, 1-9.
- Huse, E. F. *Organization development and change* (2nd ed.). New York: West Publishing Company, 1980.
- Irwin, F.W. *Intentional behavior and motivation: A cognitive view*. Philadelphia: Lippincott, 1971.
- Kraut, A. Developing managerial skills via modeling techniques: Some positive research findings. *Personnel Psychology*, 1976, 29, 325-370.
- Latham, G., & Saari, L. M. Application of social-learning theory to training supervisors through behavioral modeling. *Journal of Applied Psychology*, 1979, 64, 239-246.
- Likert, R. *The human organization: Its management and values*. New York: McGraw-Hill, 1967.
- McGregor, D. *The human side of enterprise*. New York: McGraw-Hill, 1960.
- Meichenbaum, D. H. Examination of model characteristics in reducing avoidance behavior. *Journal of Personality and Social Psychology*, 1971, 17, 298-307.
- Mischel, W. Toward a cognitive social learning reconceptualization of personality. *Psychological Review*, 1973, 80, 252-283.
- Porras, J., & Berg, P. The impact of organization development. *Academy of Management Review*, 1978, 3, 249-266.
- Porter, L. W., & Lawler, E. E. *Managerial attitudes and performance*. Homewood, Ill.: Irwin-Dorsey, 1968.
- Rogers, C., & Farson, R. *Active listening*. Chicago: Industrial Relations Center, University of Chicago, 1955.
- Vroom, V. H. *Work and motivation*. New York: Wiley, 1964.